



2-Methoxy-1,4-Naphthoquinone (MNQ) suppresses the invasion and migration of a human metastatic breast cancer cell line (MDA-MB-231)

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ABSTRACT

Metastasis contributes to the escalating mortality rate among cancer patients worldwide. The search for novel and more effective anti-metastatic agent is crucial owing to the lack of anticancer drugs that can successfully combat metastasis. Hence, this study aims to examine the effects of 2-Methoxy-1,4-Naphthoquinone (MNQ) towards the metastasis of MDA-MB-231 cells. In invasion assays, the number of cells permeating across a Matrigel barrier was found to be decreased in a dose-dependent manner upon treatment with MNQ (0–7.5 μ M). In wound-healing migration assays, MNQ exhibited dose-dependent inhibition of cell migration in which significant reduction in the zone of closure was observed as compared to untreated controls. Furthermore, the proteolytic activity of a pivotal metastatic mediator, matrix metalloproteinase-9 (MMP-9) was also downregulated by MNQ as determined by gelatin zymography. This study reports for the first time, the ability of MNQ to inhibit the invasion and migration characteristics of a highly metastatic MDA-MB-231 cancer cell line.

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1. Introduction

2-Methoxy-1,4-Naphthoquinone (MNQ) (Fig. 1) isolated from a local Malaysian medicinal plant, *Impatiens Balsamina* Linn., was previously shown to be cytotoxic against a panel of human cancer cell lines (Teng, 2010). A subsequent proteomic study by Tan (2011) revealed that MNQ induced differential expressions of proteins that are implicated in the antioxidative, stress response and detoxification pathways; thereby triggering apoptosis and cell cycle arrest in a human leukemic cell line, K562. The observed cytotoxicity of MNQ is in agreement with other works that report the cell-killing effect of MNQ on human liver and gastric cancer cell lines (Ding et al., 2008; Wang and Lin, 2012). Mori et al. (2011) further demonstrated that MNQ inhibited the Wnt signaling pathway which is crucial for cancer progression. Despite the growing number of reports on its anticancer effects, the anti-metastatic potential of MNQ has not been documented.

Globally, metastasis accounts for more than 90% of cancer-related death (Sethi and Kang, 2011). Metastatic processes in general,

involve the invasion and degradation of the basement membrane barrier by cancer cells via secretion of proteolytic enzymes such as the matrix metalloproteinases (MMPs). MMP-2 and -9 in particular, play critical roles in remodeling the extracellular matrix (ECM) by digesting various types of ECM proteins in the basement membrane (Lynch and Matrisian, 2002). Both MMPs differ in terms of their transcriptional regulations. Unlike MMP-2 which is usually constitutively expressed, the expression of MMP-9 is highly inducible by various growth factors, cytokines and other stimulatory signals. This difference can be explained by the presence of inducible promoter elements such as the activator protein-1 (AP-1) binding site which is seen in MMP-9 but not in MMP-2 (Yan and Boyd, 2007). For this reason, it has been suggested that the regulation of MMP-9 expression is a feasible approach for the development of anti-metastatic drugs (Zhang et al., 2006; Ling et al., 2011).

As a result of the proteolytic action of MMPs and other proteases, the remodeled ECMs assist the passage and the migration of the cancer cells. This in turn, facilitates their dissemination to the secondary target organs via the blood and lymph circulatory systems. The cancer cells will then further proliferate, differentiate and survive after successfully colonizing their new-found environment (Nabeshima et al., 2002; Geiger and Peeper, 2009). Hence, interference of these processes represents promising intervention strategies in anti-metastatic therapy (Valastyan and Weinberg, 2011).

Abbreviations: MNQ, 2-Methoxy-1,4-Naphthoquinone; AP-1, activator protein-1; ECM, extracellular matrix; MMP(s), matrix metalloproteinase(s); VEGF, vascular endothelial growth factor.

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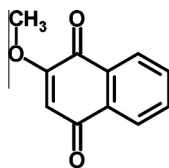


Fig. 1. 2-Methoxy-1,4-Naphthoquinone (MNQ).

As there is a lack of drugs that can successfully control metastatic progression, the use of naturally-derived phytochemicals to suppress tumor metastasis has gained immense importance (Jiang and Liu, 2011; Weber, 2013). Bioactive phytochemicals are increasingly being reported to inhibit a variety of stages in the metastatic cascade by targeting proteins and cellular signaling pathways that govern metastasis (Shu et al., 2010; Weng and Yen, 2012). Thus far, the effects of MNQ towards tumor metastasis have not been documented. In this study, we investigated the effects of MNQ at sub-toxic concentrations (2.5, 5 and 7.5 μM) on the invasion and migration of a human metastatic breast cancer cell line, MDA-MB-231. Moreover, we also examined the modulatory effect of MNQ at similar dosages towards the activity of secreted MMP-9.

2. Materials and methods

2.1. Chemicals and reagents

2-Methoxy-1,4-Naphthoquinone (98% pure), Gelatin, RPMI 1640 medium, DMSO and 0.4% Trypan Blue reagent were purchased from Sigma–Aldrich (St. Louis, MO, USA). 0.25% Trypsin/1 mM EDTA was purchased from GIBCO-BRL (Grand Island, NY, USA). Fetal bovine serum (FBS) was purchased from JRS Scientific (Woodland, CA, USA). Phosphate buffered saline (PBS) tablets were purchased from Oxoid Limited (United Kingdom). Matrigel was purchased from BD Biosciences (Bedford, MA, USA). Methanol, ethanol, acetic acid and crystal violet were purchased from Merck (Darmstadt, Germany).

2.2. Cell culture

Human breast adenocarcinoma cell line MDA-MB-231 was obtained from American Type Culture Collection (ATCC), USA. Cells were maintained in RPMI 1640 medium supplemented with 10% FBS at 37 °C, 5% CO₂. Cells were detached through incubation with trypsin/EDTA and subcultured for every two days.

2.3. Cell viability assay

MDA-MB-231 cells were seeded at 3×10^5 per 60 mm dish for 24 h. The next day, the cells were incubated in fresh medium containing different concentrations of MNQ (0, 0.625, 1.25, 2.5, 5, 10 and 20 μM) for 72 h. The final concentration of DMSO as vehicle control was less than 0.1% (v/v). After 72 h, the cell culture supernatant was removed and saved. Cells were detached via trypsinization and were pooled together with the floating cells from supernatant. The pooled cells were centrifuged at 1500 rpm for 5 min and the resultant pellets were resuspended in fresh culture medium. Trypan Blue Exclusion assay was then performed in which the number of viable and dead cells was counted using a hemocytometer. The IC₂₀ value or the concentration of MNQ that inhibited 20% cell viability was determined. The subtoxic concentrations of MNQ are defined as the concentrations that resulted in $\geq 80\%$ cell survival. Hence, IC₂₀ value or lower was used for subsequent studies.

2.4. Matrigel invasion assay

The ability of MDA-MB-231 cells to invade and pass through the Matrigel-coated membrane filter (which serves as a reconstituted basement membrane *in vitro*) was measured in a Boyden chamber assay. Each 8 μm PET membrane of the cell culture insert (BD Biosciences, USA) was first coated with 0.4 mg/mL of Matrigel for 1 h. Unbound material was aspirated and after gently rinsing with serum-free medium, the Matrigel-coated membrane was placed onto the 24-well companion plate (BD Biosciences, USA). 4×10^4 cells were plated into each upper chamber in serum-free medium with or without various concentrations of MNQ (2.5, 5 and 7.5 μM). The lower chamber was filled with 10% FBS medium which served as chemoattractant. The entire chamber was incubated for 24 h at 37 °C, 5% CO₂. After incubation, the non-invading cells were carefully removed from the upper surface of the membrane using cotton swab. The cells invading across the Matrigel to the lower surface of the membrane were fixed with 100% Methanol and stained with 0.5% Crystal Violet in 20% Methanol. The invading cells were photographed using Nikon Eclipse Ti-E inverted microscope (Nikon Instruments Inc., USA) at 100X magnification and were counted from six randomly chosen fields. Data were expressed as the percentage of invasion of treated cells as compared to the untreated cells.

2.5. Wound-healing migration assay

3×10^5 cells were plated into 60 mm dishes for 48 h to achieve 80% confluency. The wound on the confluent monolayer was created by scratching the monolayer using the yellow Gilson d200 pipette-tip. Cell debris was removed by rinsing the scratched monolayer with PBS. The cells were then incubated in 5% FBS medium with or without various concentrations of MNQ (2.5, 5 and 7.5 μM). The cells were photographed at 0, 24 and 48 h at 100X magnification and the wound area at these time points was measured using NIS Elements AR 3.2 imaging software (Nikon Instruments Inc.). The relative cell migration of each sample was calculated from the area difference in unclosed wound region measured between 0 and 48 h as compared to those of untreated control.

2.6. Gelatin zymography

3×10^5 cells were plated into 60 mm dishes for 48 h. Cells were then washed with PBS and incubated in serum-free medium with or without various concentrations of MNQ (2.5, 5 and 7.5 μM) for 24 h. The conditioned serum-free medium was collected and centrifuged to remove cells and debris. The clarified supernatant was subjected to 10-fold concentration using ethanol precipitation as described by Albini et al. (1995). After measuring the protein concentration of each sample using Bradford assay (Sigma, USA), gelatin zymography was performed as previously reported by Toth et al. (2012). Briefly, the protein sample (3 $\mu\text{g}/\text{lane}$) was prepared in SDS sample buffer without boiling and addition of reducing agent. Next, the protein sample was electrophoresed on a 10% SDS-PAGE gel co-polymerized with 0.1% w/v gelatin. The resulting gels were incubated in renaturing solution (2.5% v/v Triton X-100) for 30 min, followed by one-time rinsing with double-distilled water, and incubation in developing buffer (50 mM Tris–HCl pH 7.8, 200 mM NaCl, 5 mM CaCl₂, 0.2% v/v Brij 35) for another 30 min at room temperature. The gels were then incubated with fresh developing buffer for 16 h at 37 °C. The gel was stained with Coomassie Brilliant Blue R-250 solution for 1 h, followed by destaining for 2 h using destaining solution (10% methanol, 5% acetic acid). Enzyme-digested regions were observed as clear bands against a dark blue background. Gels were scanned using the

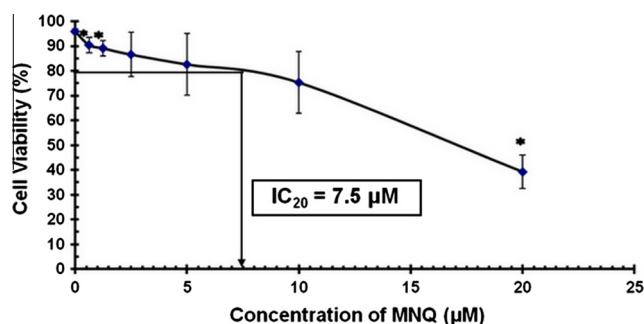


Fig. 2. The cell viability of MDA-MB-231 cells upon treatment with MNQ for 72 h. The subtoxic concentration range of these compounds that resulted in $\geq 80\%$ survival was used for subsequent studies. Data are expressed as means $\pm$ standard deviation for three independent experiments with triplicate each. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; significant difference from untreated controls.

Bio-Rad GS-800 Calibrated Imaging Densitometer and the subsequent zymographic band intensities were quantified using QuantityOne software (Bio-Rad, Hercules, CA).

2.7. Statistical analysis

All data were expressed as mean $\pm$ standard deviation (S.D.). The difference between the treated samples and the untreated controls was analyzed by Student's *t*-test in which $p < 0.05$ was considered as significant.

3. Results

3.1. Effect of MNQ towards MDA-MB-231 cell viability

The effect of MNQ on MDA-MB-231 cell viability was determined using Trypan Blue Exclusion Assay. As shown in Fig. 2, the viability of MDA-MB-231 cells decreased in a dose-dependent manner upon treatment with MNQ for 72 h. From the viability curve plotted, the IC_{20} value of MNQ was determined to be $7.5 \mu M$. For the subsequent assays, the concentration range of MNQ tested was 2.5, 5 and $7.5 \mu M$. These concentrations were

deemed to be subtoxic to MDA-MB-231 cells as $\geq 80\%$ cell survival was still maintained.

3.2. MNQ inhibits the invasion and migration of MDA-MB-231 cells

Two assays were performed to evaluate whether MNQ possesses the inhibitory effects against tumor invasion and migration. In the Matrigel Invasion Assay, MNQ at subtoxic concentrations (2.5, 5 and $7.5 \mu M$) inhibited MDA-MB-231 cell invasion across the reconstituted basement membrane *in vitro*. As shown in Fig. 3, the number of cells found on the lower side of Matrigel-coated membrane filter was observed to decrease in a dose-dependent manner. Fifty-nine percentage inhibition of cell invasion was recorded upon treatment with $7.5 \mu M$ of MNQ. Furthermore, we demonstrated that MNQ was able to retard the migration of MDA-MB-231 cells in the Wound-Healing Migration Assay. The ability of the cells to move and to repopulate the scratched wound region was suppressed by MNQ also in a dose-dependent manner (Fig. 4). At $7.5 \mu M$, MNQ reduced wound closure by 46% relative to untreated controls.

3.3. MNQ downregulates the proteolytic activity of MMP-9

Tumor invasion relies heavily on the action of extracellularly-secreted proteases such as MMPs to degrade the basement membrane barrier and to assist tumor escape and migration. Hence, gelatin zymography was employed to assess the effects of MNQ on the activity of secreted MMP-9, which is a key ECM-degrading molecule. As shown in Fig. 5, the appearance and the intensity of the clear bands which reflected the proteolytic activity of MMP-9 were reduced upon exposure to MNQ at different subtoxic concentrations. At $7.5 \mu M$, MNQ reduced the activity of MMP-9 by 70% as compared to the untreated controls.

4. Discussion

MNQ, a naturally occurring phytochemical in the plant *Impatiens Balsamina* Linn., has been reported to possess cytotoxic effects against different cancer cells. Our results showed that MNQ exhibited dose-dependent toxicity towards MDA-MB-231 cells. This

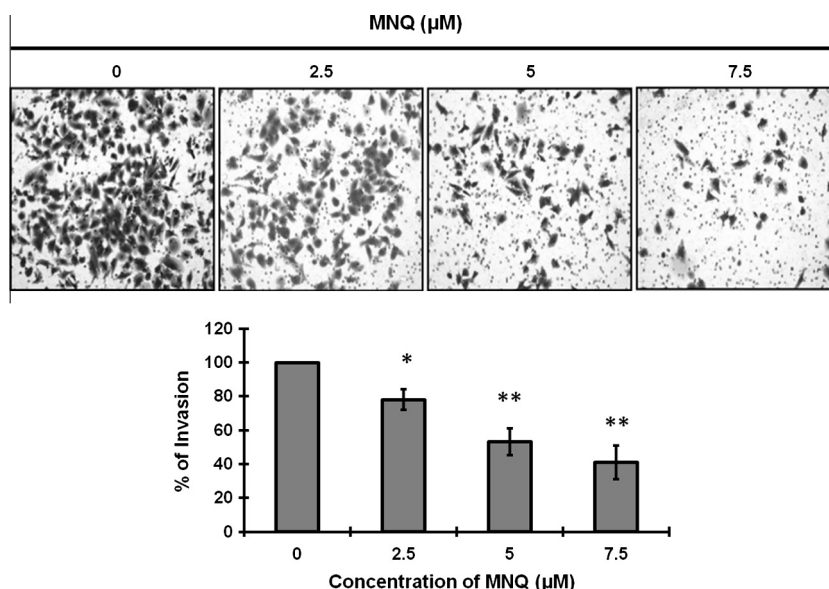


Fig. 3. The effects of MNQ on the invasion of MDA-MB-231 cells towards the Matrigel-coated filters after 24 h. All the cell images were captured at $100\times$ magnification. Data are expressed as means $\pm$ standard deviation for three independent experiments with triplicate each. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; significant difference from untreated controls.

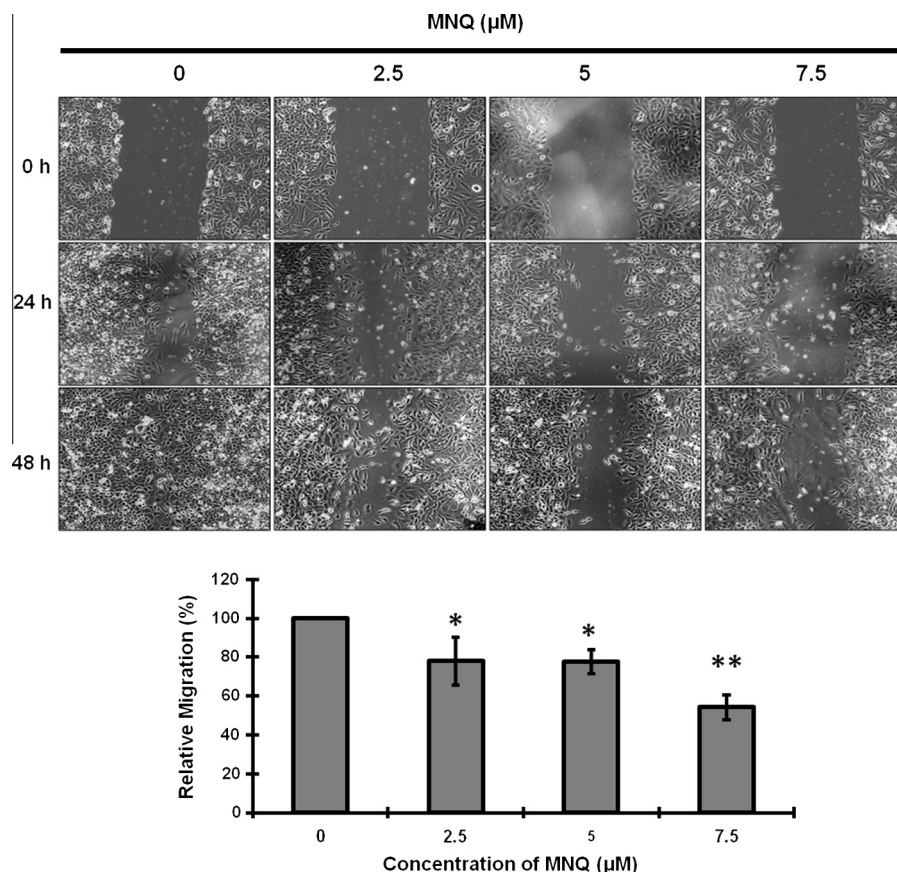


Fig. 4. The effects of MNQ towards the migration of MDA-MB-231 cells. All the cell images were captured at 100 × magnification. The relative migration of each sample was calculated from the difference in area of unclosed wound region measured between 0 and 48 h as compared to those of untreated control. Data are expressed as means ± standard deviation for three independent experiments with triplicate each. * $P < 0.05$; ** $P < 0.01$; significant difference from untreated controls.

observation is in agreement with the cytotoxic property of MNQ reported in other findings (Ding et al., 2008; Teng, 2010; Wang and Lin, 2012).

Nonetheless, little is known about its anti-metastatic potential. Hence, the invasion and migration potential of MDA-MB-231 cells were challenged with treatment of MNQ at subtoxic concentrations ($\leq \text{IC}_{20}$). This concentration range (2.5, 5 and 7.5 μM) was used in order to avoid false positive results contributed by the cell-killing effects of MNQ at higher doses. We demonstrated for the first time that MNQ, at subtoxic concentrations could inhibit the metastasis of MDA-MB-231 cells in a dose-dependent manner. MNQ at its highest subtoxic concentration, 7.5 μM blocked cell invasion by 59%, thus reducing the number of cells permeating through the Matrigel barrier. At this similar dose, MNQ was shown to reduce wound closure by 46% which indicates an inhibition of cell migration.

Thus far, the detailed mechanism by which MNQ exerts inhibitory effects against the metastasis of MDA-MB-231 cells is still unclear. Prior studies on the closely-related 1,4-Naphthoquinones suggest that the anti-metastatic properties of this family of compounds can be attributed to the downregulation of the activities of matrix metalloproteinase (MMP) enzymes. Plumbagin for example was reported to decrease the activity of MMP-2 in human liver (HepG2) and lung (A549) cancer cell lines. Shikonin on the other hand was shown to reduce the MMP-9 activity in human adenoid cystic carcinoma ACC-M cells (Shih et al., 2009; Shieh et al., 2010; Min et al., 2011). High expression of MMPs strongly correlates with the invasive and aggressive behavior of tumors. These MMPs are secreted to degrade the ECM proteins in the basement membrane

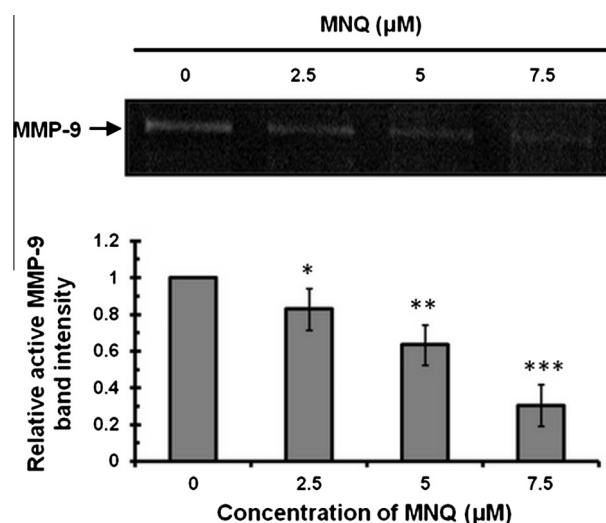


Fig. 5. The effects of MNQ towards the proteolytic activity of MMP-9 as measured by gelatin zymography. Densitometric analysis of the clear bands on the zymograms was performed using the QuantityOne software. Data are expressed as means ± standard deviation for minimum three independent experiments. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; significant difference from untreated controls.

barrier which eventually promotes the invasion and migration behavior of the tumor cells (Lynch and Matrisian, 2002; Kessenbrock et al., 2010).

In the present study, we demonstrated that MMP-9 activity in the conditioned serum-free medium of MDA-MB-231 cells was significantly downregulated by MNQ. At 7.5 μ M, MNQ decreased MMP-9 activity by 75%. This suggests that the inhibition of cell invasion and migration by MNQ could be mediated by the reduction in the proteolytic action of MMP-9. MMP-9 is pivotal in the invasion process by degrading the key ECM components in the basement membrane barrier such as Type IV collagen (Bjorklund and Koivunen, 2005). Moreover it was reported that MMP-9 interacts with integrin α v β 3 to promote the migration of breast cancer cells (Rolli et al., 2003). Interestingly, MMP-9 was also shown to be a pro-angiogenic switch by regulating the bioavailability of vascular endothelial growth factor (VEGF), the most potent inducer of tumor angiogenesis (Hua et al., 2011).

In conclusion, our data indicated for the first time that MNQ could inhibit migration and invasion of a metastatic cell line at least via the downregulation of MMP-9 activity. Further studies on its anti-metastatic potential are currently in progress to elucidate its mode of action, particularly in the regulation of MMP-9. It may also be instructive to compare the activity of naphthoquinone analogues via a Structure-Activity Relationship (SAR) study which may highlight the contribution of various functional groups towards their anti-metastatic potency.

Conflict of Interest

The authors report no conflict of interest.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.tiv.2013.11.008>.

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